

IE4-DBL: Lab 2 assignment

Deadline: Sunday 11:59 PM before your lab session

What to upload: One PDF file per team, with:

- your names
- your matriculation numbers
- solutions for all assignments

Where to upload? Moodle room, "Submission Lab 2"

In preparation of the lab, you must prepare the assignments and submit your solution in the Moodle room for this course. The submission must be a PDF file. During the lab, each team member must be able to present all parts of the solution.

You can use a drawing tool of your choice to create the diagrams or just draw the diagrams on a piece of paper. You can also use any DBMS to develop / test your SQL scripts.

If you have questions or issues, please ask them in the Moodle room.

Assignment 1: Bar owner

The owner of several bars wants to supervise his business:

- First, each bar has a unique name and offers a variety of different drinks. A bar is located at a specific address.
- Different bars may serve the same drinks but at different prices.
- The drinks are made of different ingredients. Since the drinks are sometimes improved, these drinks have the same name but require some other kind of distinction.
- The only employees are bartenders that are only employed in one bar. Each bar needs several bartenders to fulfil the job.
- Even though a bar may have different bartenders, these bartenders work only with a fixed partner (another bartender). Some bartenders don't have a partner and thus work alone. Bartenders have a name, a social security number, a salary, an address and a phone number for contact purposes.

Assignments:

1. Draw an entity-relationship-model according to these requirements, including entities, relationships and attributes. Mark the key attributes of each entity. Add for each relationship the cardinality and participation type.
2. Convert the ERM into a relational schema.
3. Write SQL-statements that create the corresponding tables. Come up with reasonable constraints and datatypes for the fields of the tables. Insert some data into the tables.

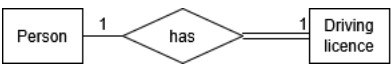
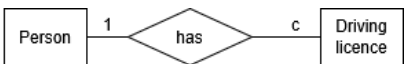
Assignment 2: Journals

Persons, described by their name, SSN, and address, subscribe to various journals. Each journal, identified by a title, has a set of numbered volumes identified by a unique name and a unique ISBN. Every volume has a set of numbered articles. Subscribers have an initial subscription date and a termination date for each journal to which they subscribe.

Draw an ERM and an ER to cover all aspects described above.

Assignment 3: Relationship cardinality notation

Draw these relationships in Chen and mc notation, as in the example.

	Chen	MC
Example: person – has – driving licence		
Author – writes - Book (e.g. in a library information system) Note: you are only considered to be an author if you have written a book.		
Author – writes – Debut novel		
Customer – places - Order (e.g. on ebay) Note: you create your customer account before you actually place an order		
House – has – door Note: if it hasn't a door, we don't call it a house		

Assignment 4: Normalization for a primary school student's list

A primary school keeps lists of all students and their grades. But elegant and robust database design is not yet a primary school topic: their table is clearly not in 3NF.

Bring the following table to 3NF.

It's sufficient to provide only the 3NF form, but if you're into some extra exercise, you can also provide a 1NF and 2NF solution.

<u>Student number</u>	<u>Student name</u>	<u>Father / Mother / Legal guardian</u>	<u>Course</u>	<u>Class teacher</u>	<u>Class</u>	<u>Grade</u>
375428	Yuna Tanaka	Haruko Tanaka, Kenji Tanaka	Mathematics 1	Mr Schneider	1A	A
234982	Lucas Martins	Renata Martins, Paulo Martins	Mathematics 1	Mr Schneider	1A	A-
234812	Tariq Hassan	Sara Hassan, Ahmed Hassan	Mathematics 1	Mr Schneider	1A	B+
375428	Yuna Tanaka	Haruko Tanaka, Kenji Tanaka	Reading 1	Mr Schneider	1A	A
234982	Lucas Martins	Renata Martins, Paulo Martins	Reading 1	Mr Schneider	1A	A
234812	Tariq Hassan	Sara Hassan, Ahmed Hassan	Reading 1	Mr Schneider	1A	A
930485	Amira Rahman	Leila Rahman, Hassan Rahman	Mathematics 1	Mrs. O'Connor	1B	B+
234980	Emma Johansson	Sofia Johansson	Mathematics 1	Mrs. O'Connor	1B	A-
120398	Maya Patel	Priya Patel, Rakesh Patel	Mathematics 1	Mrs. O'Connor	1B	B+
930485	Amira Rahman	Leila Rahman, Hassan Rahman	Reading 1	Mrs. O'Connor	1B	B
234980	Emma Johansson	Sofia Johansson	Reading 1	Mrs. O'Connor	1B	B+
120398	Maya Patel	Priya Patel, Rakesh Patel	Reading 1	Mrs. O'Connor	1B	B-
234981	Samuel Okafor	Ada Okafor, Chinedu Okafor	Mathematics 4	Mr. Schubert	4A	C+
239408	Omar Al-Fayed	Lucía Martínez, Carlos Martínez	Mathematics 4	Mr. Schubert	4A	A-
234981	Samuel Okafor	Ada Okafor, Chinedu Okafor	English 4	Mr. Schubert	4A	B+
239408	Omar Al-Fayed	Lucía Martínez, Carlos Martínez	English 4	Mr. Schubert	4A	B-
234981	Samuel Okafor	Ada Okafor, Chinedu Okafor	Art 4	Mr. Schubert	4A	C+
239408	Omar Al-Fayed	Lucía Martínez, Carlos Martínez	Art 4	Mr. Schubert	4A	A-
123456	Lucas Martins	Franco Martins	English 4	Mr. Schubert	4A	B
123456	Lucas Martins	Franco Martins	Art 4	Mr. Schubert	4A	A
123456	Lucas Martins	Franco Martins	Mathematics 4	Mr. Schubert	4A	B-

Assignment 5: DQL basics for a recipe database

The following tables show some example entries from a recipe database.

Write an SQL statement for the following queries:

- Get a table with the names of all recipes
- Get a table with the name and cuisine type of all recipes that are easy
- Get a table with all ingredients and their amount for Spaghetti Carbonara
- Get a table with all ingredients and their amount for recipes that are easy. Also include the name and the cuisine type for all recipes that are easy.
- Get a table of all recipe names that go without onions

Bonus question: If you are allowed to change one thing to the table “ingredient2recipe”, what would you change?

Table “recipes”:

Recipe_ID	Recipe_Name	Difficulty	Date_Created	Cuisine_Type
1	Spaghetti Carbonara	Medium	12.03.2024	Italian
2	Chicken Tikka Masala	Hard	22.05.2024	Indian
3	Greek Salad	Easy	08.07.2024	Mediterranean
4	Sushi Rolls	Hard	15.09.2024	Japanese
5	Chocolate Chip Cookies	Easy	10.10.2024	American

Table “ingredient2recipe”:

Recipe_ID	Ingredient_Name	Amount
1	Spaghetti	200 g
1	Eggs	2
1	Parmesan cheese	50 g
1	Bacon	100 g
2	Chicken breast	300 g
2	Yogurt	150 g
2	Tikka masala paste	2 tbsp
2	Onion	1
3	Tomato	2
3	Cucumber	1
3	Feta cheese	100 g
3	Olive oil	2 tbsp
3	Onion	2
4	Sushi rice	200 g
4	Nori sheets	4
4	Cucumber	1
4	Salmon	100 g
5	Flour	250 g
5	Sugar	100 g
5	Chocolate chips	80 g
5	Butter	100 g